COMPUTER PROGRAMMING

LAB 4

LAB TITLE

Switch Statement, Break and Continue in C++

1. Objective

The objective of this lab is to understand how the **switch statement** works in C++. In this lab, students learn how to select one option from many choices using switch. We also learn how the **break statement** controls the flow of the program and how the **continue statement** works inside loops. This lab also helps students create simple **menu-driven programs**.

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2. Theory

In C++, sometimes a program needs to choose one option from many choices. For this purpose, the **switch statement** is used. It is often used instead of many **if-else statements** when comparing one variable with different values.

Inside a switch statement, different options are written as **cases**. The program checks the value of the variable and runs the matching case.

The **break statement** is used to stop the execution of a case. If break is not used, the program will continue running the next cases.

The **default statement** runs when none of the cases match the user input.

The **continue statement** is mainly used in loops. It skips the current iteration and moves to the next iteration of the loop.

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3. Software

- DEV-C++ IDE
- C++ Compiler (included in DEV-C++)

Program 1: Day Name Using Switch

Algorithm

1. Start the program
2. Declare an integer variable for the day number
3. Ask the user to enter a number from **1 to 7**
4. Use a **switch statement** to check the number
5. Display the corresponding day name
6. End the program

Program 2: Grade Evaluation Using Switch

Algorithm

1. Start the program
2. Declare a character variable for grade
3. Ask the user to enter a grade
4. Use a **switch statement** to check the grade
5. Display the message according to the grade
6. End the program

Program 3: Unit Conversion Menu Using Switch

Algorithm

1. Start the program
2. Display a menu with different conversion options
3. Take the user's choice
4. Ask the user to enter a value
5. Perform the selected conversion
6. Display the result
7. End the program

Conclusion

In this lab, we learned how to use the **switch statement** for making multiple decisions in C++. We also understood the importance of the **break statement** in controlling the flow of the program. This lab helped us create **simple menu-based programs** and improved our understanding of conditional structures in C++.